Data Analysis Portfolio

Data Analysis and Visualization using Python

Student: Shravani Chaudhari

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What is Data Analysis?

Data analysis is the process of examining, cleaning, and transforming datasets to discover useful information, draw conclusions, and support decision-making. Organizations across all industries rely on data analysis to make informed strategic choices.

This portfolio demonstrates practical data analysis skills using real-world datasets and Python programming tools including Pandas, NumPy, Matplotlib, and Seaborn for data manipulation and visualization.

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Project Objectives

Real-World Datasets

Work with authentic datasets from sales, education, climate, finance, and social media domains

Data Cleaning

Master techniques for handling missing values, removing duplicates, and standardizing data formats

Exploratory Analysis

Conduct comprehensive EDA to understand data distributions, correlations, and patterns

Visual Storytelling

Create compelling visualizations that communicate insights clearly and effectively

Generate Insights

Transform raw data into actionable business and strategic recommendations

Professional Portfolio

Build a showcase of skills demonstrating readiness for data analyst roles

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Tools and Technologies

Python

Core programming language for data analysis, known for its simplicity and powerful libraries

Pandas

Data manipulation library for handling structured data with powerful data frame operations

NumPy

Fundamental package for numerical computing with efficient array operations and mathematical functions

Matplotlib

Foundation for creating static, animated, and interactive visualizations in Python

Seaborn

Statistical visualization library built on Matplotlib, creating attractive charts with less code

VS Code

Lightweight code editor with excellent Python support, debugging tools, and extensions

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Project Workflow

The data analysis process follows a structured methodology from raw data to actionable insights.

Data Collection

Gather datasets from reliable sources and APIs

Data Cleaning

Handle missing values, remove duplicates, fix inconsistencies

Exploratory Analysis

Calculate statistics, identify patterns and correlations

Data Visualization

Create charts and dashboards to communicate findings

Insight Generation

Transform data into strategic recommendations

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Sales Data Analysis

Project Overview

Analyzed a comprehensive supermarket sales dataset to uncover business insights and optimize operations.

Key Analysis Areas

- Identified best-selling product categories and items
- Analyzed daily, weekly, and monthly sales trends
- Studied customer purchasing behavior and patterns
- Examined regional performance variations

Visualizations Created

Bar charts showing top products, line graphs displaying sales trends over time, and heatmaps revealing peak purchasing periods.

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Student Performance Analysis

Research Questions

- What factors most strongly influence student test scores?
- Are there performance differences between genders?
- How does parental education level impact outcomes?
- What is the effect of test preparation courses?

Analysis Approach

Conducted comparative analysis using statistical methods and created visualizations including bar charts, box plots, and correlation matrices to reveal relationships between variables.

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Weather and Climate Analysis

Temperature Patterns

Analyzed historical temperature records to identify seasonal cycles and long-term trends across different geographic regions

Precipitation Analysis

Examined rainfall data to understand seasonal distribution, extreme weather events, and drought patterns

Climate Change Indicators

Studied decade-long datasets to detect shifts in weather patterns and potential climate change impacts

Created line graphs showing temperature trends, rainfall distribution charts, and seasonal pattern visualizations using time series analysis.

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Financial Market Analysis

India VIX Volatility Study

Analyzed India VIX (Volatility Index) dataset to understand market sentiment and risk levels in the Indian stock market.

Key Findings

- Identified periods of high financial uncertainty
- Analyzed relationship between volatility and market movements
- Detected seasonal patterns in market volatility

Visualizations

Created line charts showing volatility trends over time, histograms displaying volatility distribution, and correlation analysis with market indices.

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Social Media Sentiment Analysis

Project Scope

Analyzed social media posts and comments to understand public opinion and engagement patterns.

Analysis Components

- Classified sentiment as positive, negative, or neutral
- Tracked sentiment changes over time
- Identified key topics driving engagement
- Analyzed correlation between sentiment and user engagement

Visualizations

Pie charts showing sentiment distribution, line graphs displaying sentiment trends, and word clouds highlighting frequently mentioned topics.

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